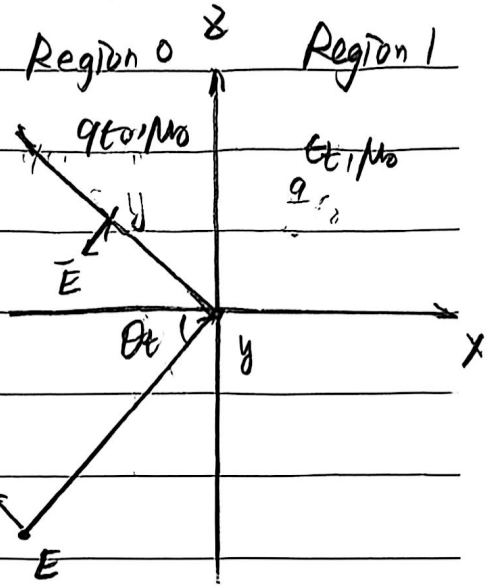
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P2



(1) $\vec{E_i}=(-\hat{x}+\sqrt{2}\hat{y}+\hat{z})E_0e^{j(k_xx+k_zz)}$

$\vec{k}=k_x\hat{x}+k_z\hat{z}$

$\vec{k}\cdot\vec{E_i}=0\Rightarrow -k_x+k_z=0$

故 $k_x=k_z$ 故 $\theta_t=45°$

(2) $k_z=\pi m^{-1}$, $c=3\times10^8 m/s$

$k^2=k_x^2+k_z^2=\omega^2\mu\varepsilon$

$\Rightarrow f=\frac{c}{3\sqrt{2}}=0.07GHz \quad \lambda=\frac{c}{f}=3m$

(3) $\sin\theta_t=\frac{k_x}{k_0}=\left|\frac{E_t}{\eta E_0}\right|=\frac{\sqrt{2}}{2}$

$\Rightarrow E_t=\frac{\eta}{2}E_0$

$\vec{E_i}=(-\hat{x}+\hat{z})E_0e^{j(k_xx+k_zz)}+\sqrt{2}\hat{y}E_0e^{j(k_xx+k_zz)}$ 右旋圆极化

$\rho_{01}^{TM}=\frac{\eta_0k_{tx}}{\frac{\eta}{2}k_x}=0 \quad R_{01}^{TM}=1$

$\rho_{01}^{TE}=\frac{\mu_0k_{tx}}{\mu_0k_x}=0 \quad R_{01}^{TE}=1$

$\vec{k_r}=-k_x\hat{x}+k_z\hat{z}$

$\vec{E_r}=(-\hat{x}-\hat{z})E_0e^{j(-k_xx+k_zz)}-\sqrt{2}\hat{y}E_0e^{j(k_xx+k_zz)}$ 左旋圆极化

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